

Expo and Defense

Project: Helpful Classroom Sorter

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Improve One Cause

Without opening the old lesson, explain one key idea from Improve One Cause. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Test Normal Cases

Without opening the old lesson, explain one key idea from Test Normal Cases. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Choose Inputs and Outputs

Without opening the old lesson, explain one key idea from Choose Inputs and Outputs. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Keep the final evaluation record for Evaluating and explaining your project. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Evaluating and explaining your project. Proposed explanation: “A polished presentation proves the system works”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Judge a different project independently

A fictional library adviser and its manual baseline face the same 6 reserved cases. The candidate has 4 full passes; the baseline has 5. One additional planned case is unrun by both. No speed data was collected. The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the evidence?

6. Solve the given case

The expo poster says always correct after six classroom passes.

Write a defensible claim.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

At the expo, a team says our sorter never fails after testing four cards. Replace the claim with an evidence-based statement and name a next test.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 36 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Present the final Classroom Sorter portfolio and complete the AI-1 certification defense.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Defend a sorter with 10/12 routine successes and 2/4 mixed-material successes. Report overall and separate results, show one failure and propose a justified next step. What claim would overstate the evidence?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Record the fixed candidate

Before final cases are revealed, record your version label, where its complete instructions are saved, the agreed input and matching rules, the manual baseline and any known limitation. Confirm that the instructions stay unchanged while the assigned catalogue is used as input.

My reasoning and evidence

13. W2 Keep the final evaluation record

Use a separate sheet for the teacher's reserved cases. For every case record its ID and inputs, catalogue and candidate labels, both methods' observed outputs and reasons, and results after checking. Preserve failure traces. Here summarise each method's full passes over attempted cases and list unrun cases. State what the comparison supports.

My reasoning and evidence

14. W3 Prepare your evidence defence

Write a two-minute explanation of the problem, one trace, your actual comparison, a limitation and a next action. Explain why this is a written-rule design rather than a model trained on examples. State which personal information is unnecessary and who decides whether to act.

My reasoning and evidence

15. W4 Judge a different project independently

A fictional library adviser and its manual baseline face the same 6 reserved cases. The candidate has 4 full passes; the baseline has 5. One additional planned case is unrun by both. No speed data was collected. The team says “ours is faster and more reliable everywhere” and repairs it after seeing the failures. Explain what the counts support, two unsupported claims and how to label a rerun of the revealed cases.

My reasoning and evidence

16. W5 Write your final conclusion

Use your own records to complete: Under these conditions, our saved version showed... The baseline showed... We still do not know... Our next action is... Include one privacy or human-control choice. If a case failed, explain its consequence without deleting the result.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Expo and Defense

A fictional paper sorter team prepares an exhibition evidence board.

Both prototype and manual baseline attempt the same 10 reserved cases. Prototype passes 8; baseline passes 9.

Two additional planned cases are unrun. No timing or pupil-satisfaction measures were collected.

Shared reserved-case comparison

passes out of 10 attempted

Prototype



Manual baseline



Bar length shows the supplied value; the marked baseline is zero.

Expo and Defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Describe contract

Show the task, inputs, rule and reserved case selection.



2. Show results

Prototype $8/10=80\%$; baseline $9/10=90\%$ on attempted cases.



3. Keep incomplete work

Two unrun cases remain visible, without pass/fail labels.



4. Defend conclusion

Prototype did not beat baseline on this measured outcome; explain its failures.

CHANGE ONE THING

Someone adds the two unrun cases to the denominator as failures.

Explain the changed result using the matching lesson.

Write one defensible success statement and one claim the evidence cannot support.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Present the Helpful Classroom Sorter's versions, test records, baseline, failures and next investigation as an evidence portfolio.

19. Project portfolio milestone

Project: Helpful Classroom Sorter

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Present the Helpful Classroom Sorter's versions, test records, baseline, failures and next investigation as an evidence portfolio.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional paper sorter team prepares an exhibition evidence board.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Prototype did not beat baseline on this measured outcome; explain its failures.

1. Keep incomplete work
2. Describe contract
3. Show results
4. Defend conclusion

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Defend conclusion Prototype did not beat baseline on this measured outcome; explain its failures. An exhibition should make the evidence inspectable. A polished demonstration is not a substitute for the reserved-case record. The shared cases allow a direct descriptive comparison of pass rates; missing timing data prevents a claim that the prototype is faster. Unrun cases must not be treated as successes.

Practice 2: explain the deciding evidence

Changed-case practice: Someone adds the two unrun cases to the denominator as failures. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed input or the procedure.
2. This one example guarantees correct results for every future input.
3. That reports a different measure, 8/12 completed successes out of planned cases.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That reports a different measure, 8/12 completed successes out of planned cases. It cannot be called the original attempted-case pass rate; distinguish completion from evaluated correctness.

Repair a missing building block

Review Improve One Cause (ai-1/35). Goal: Repair a located defect without changing the agreed requirements. Short practice: Explain why “our repaired version now picks something” is not enough to show success at 6.. Return to this week and explain what you changed.

Review Test Normal Cases (ai-1/33). Goal: Plan routine cases with expected results derived from the requirements. Short practice: Why is none correct for 8 but incorrect for 6? Use both capacity cards.. Return to this week and explain what you changed.

Review Choose Inputs and Outputs (ai-1/30). Goal: Turn the project outcome into precise input and output requirements. Short practice: Trace requests 10 and blank. Why should they not receive the same explanation?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
